1 Lecture 13: March 17, 2026

Lecture Overview: We introduce the Riemann integral via the machinery of partitions, upper sums, and lower sums. A partition of $[a, b]$ splits the interval into finitely many subintervals; on each one we take the supremum and infimum of a bounded function f to form upper and lower Darboux sums $U(P, f)$ and $L(P, f)$. Taking the infimum over all partitions gives the upper Riemann integral $\overline{\int}$, and the supremum gives the lower Riemann integral $\underline{\int}$. When these two agree, f is Riemann integrable.

1.1 Partitions and Darboux Sums

Section Overview: We define partitions of a closed interval $[a, b]$, the associated quantities Δx_i, M_i, m_i, and build the upper and lower Darboux sums $U(P, f)$ and $L(P, f)$ that bound the "area under the curve" from above and below.

Definition 1.1. A **partition** of $[a, b]$ is a finite set of points $P = \{x_0, x_1, \ldots, x_n\}$ such that

$$a = x_0 \leq x_1 \leq \cdots \leq x_n = b.$$

Definition 1.2. Given a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ and a bounded function $f : [a, b] \to \mathbb{R}$, define for each $i = 1, 2, \ldots, n$:

$$\begin{aligned} \Delta x_i &= x_i - x_{i-1}, \\ M_i &= \sup_{x \in [x_{i-1}, x_i]} f(x), \\ m_i &= \inf_{x \in [x_{i-1}, x_i]} f(x). \end{aligned}$$

The **upper Darboux sum** of f with respect to P is

$$U(P, f) = \sum_{i=1}^{n} M_i \, \Delta x_i,$$

and the **lower Darboux sum** of f with respect to P is

$$L(P, f) = \sum_{i=1}^{n} m_i \, \Delta x_i.$$

Remark 1.3. Since $m_i \leq M_i$ for all i, we always have $L(P, f) \leq U(P, f)$ for any partition P.

1.2 Upper and Lower Riemann Integrals

Section Overview: By taking the infimum of all upper sums and the supremum of all lower sums, we obtain the upper and lower Riemann integrals. When these coincide, f is Riemann integrable on $[a, b]$.

Definition 1.4. Let $f : [a, b] \to \mathbb{R}$ be bounded. The **upper Riemann integral** of f over $[a, b]$ is

$$\overline{\int_a^b} f\,dx = \inf_P U(P, f),$$

and the **lower Riemann integral** of f over $[a, b]$ is

$$\underline{\int_a^b} f\,dx = \sup_P L(P, f),$$

where the infimum and supremum are taken over all partitions P of $[a, b]$.

Definition 1.5. A bounded function $f : [a, b] \to \mathbb{R}$ is **Riemann integrable** on $[a, b]$, written $f \in R$ on $[a, b]$, if

$$\overline{\int_a^b} f\,dx = \underline{\int_a^b} f\,dx.$$

In this case, the common value is called the **Riemann integral** of f and is denoted

$$\int_a^b f\,dx.$$

Definition 1.6. A **measure** on a set X is a function $\mu : 2^X \to \mathbb{R}$ such that

(i) $\mu(\emptyset) = 0$,

(ii) $\mu(E) \geq 0$ for all $E \subseteq X$,

(iii) If $E_1, E_2, \ldots$ are pairwise disjoint, then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$.

Example. Examples of measures:

(i) The **Lebesgue measure** on $\mathbb{R}$ assigns to each interval its length: $\mu([a, b]) = b - a$.

(ii) The **counting measure** (discrete measure): $\mu(E) = |E|$, the number of elements in E.

(iii) The **Haar measure** on a group: a measure that is invariant under the group operation.

(iv) A **probability measure** (distribution): a measure with $\mu(X) = 1$.

(v) The **Dirac measure** at a point x_0: $\mu(E) = \begin{cases} 1 & \text{if } x_0 \in E, \\ 0 & \text{if } x_0 \notin E. \end{cases}$

Theorem 1.7. *The Riemann integral $\int_a^b f\,dx$ exists (i.e., $f \in R$ on $[a, b]$) in each of the following cases:*

(i) f is continuous on $[a, b]$,

(ii) f is monotonic on $[a, b]$,

(iii) f is bounded on $[a, b]$ with finitely many discontinuities,

(iv) $f \circ g$ where $g \in R$ on $[a, b]$.

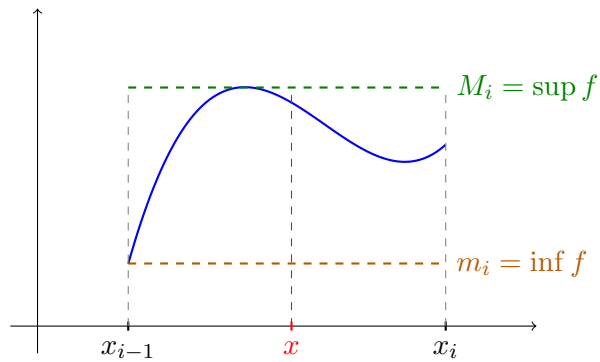
Definition 1.8. Let P be a partition of $[a, b]$. A partition P^* is a **refinement** of P if $P \subseteq P^*$.

Theorem 1.9. *If P^* is a refinement of P, then*

$$L(P, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f).$$

Proof. By induction on $|P^* \setminus P|$.

Base case: $P^* \setminus P = \{x\}$, a single point, where $x_{i-1} \leq x \leq x_i$ for some i.



Adding the single point x splits the interval $[x_{i-1}, x_i]$ into $[x_{i-1}, x]$ and $[x, x_i]$. Let

$$w_1 = \inf_{t \in [x_{i-1}, x]} f(t), \quad W_1 = \sup_{t \in [x_{i-1}, x]} f(t),$$
$$w_2 = \inf_{t \in [x, x_i]} f(t), \quad W_2 = \sup_{t \in [x, x_i]} f(t).$$

Since $[x_{i-1}, x] \subseteq [x_{i-1}, x_i]$ and $[x, x_i] \subseteq [x_{i-1}, x_i]$, we have

$$m_i \leq w_1, \quad m_i \leq w_2, \quad W_1 \leq M_i, \quad W_2 \leq M_i.$$

Now the lower sum changes only in the i-th term:

$$\begin{aligned} L(P^*, f) - L(P, f) &= w_1(x - x_{i-1}) + w_2(x_i - x) - m_i(x_i - x_{i-1}) \\ &\geq m_i(x - x_{i-1}) + m_i(x_i - x) - m_i(x_i - x_{i-1}) = 0. \end{aligned}$$

So $L(P, f) \leq L(P^*, f)$. Similarly,

$$\begin{aligned} U(P, f) - U(P^*, f) &= M_i(x_i - x_{i-1}) - W_1(x - x_{i-1}) - W_2(x_i - x) \\ &\geq M_i(x_i - x_{i-1}) - M_i(x - x_{i-1}) - M_i(x_i - x) = 0. \end{aligned}$$

So $U(P^*, f) \leq U(P, f)$. The inequality $L(P^*, f) \leq U(P^*, f)$ is immediate, giving

$$L(P, f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P, f).$$

Inductive step: If $|P^* \setminus P| = k$, pick any $x \in P^* \setminus P$ and let $P' = P \cup \{x\}$. Then $|P^* \setminus P'| = k-1$, so by the inductive hypothesis

$$L(P', f) \leq L(P^*, f) \leq U(P^*, f) \leq U(P', f),$$

and by the base case

$$L(P, f) \leq L(P', f) \quad \text{and} \quad U(P', f) \leq U(P, f).$$

Combining gives the result. $\square$

Remark 1.10. One might ask: can we induct on uncountably infinite sets? Ordinary induction only applies to $\mathbb{N}$, so it handles finite steps. This is not an issue here, since partitions are finite sets by definition, and $|P^* \setminus P| \in \mathbb{N}$. For uncountable settings, there is a more powerful tool called **transfinite induction** (induction over ordinals), but this belongs to set theory and is beyond the scope of this course.

Technique: Induction on set difference. When you need to show a property holds for a refinement $P^* \supseteq P$, induct on $|P^* \setminus P|$, the number of added points. The base case handles adding a single point — this is where the real work happens, and it typically reduces to comparing quantities on one interval versus its two sub-intervals. The inductive step is then essentially free: add one point at a time, apply the base case, and chain the inequalities. This "one point at a time" strategy appears whenever you refine a partition, mesh, or discretization.

Theorem 1.11. *For any bounded function f on $[a, b]$,*

$$\sup_P L(P, f) \leq \inf_P U(P, f).$$

That is, $\underline{\int_a^b} f\,dx \leq \overline{\int_a^b} f\,dx$.

Proof. Let P_1 and P_2 be any two partitions of $[a, b]$. Then $P_1 \cup P_2$ is a refinement of both, so by the previous theorem:

$$L(P_1, f) \leq L(P_1 \cup P_2, f) \leq U(P_1 \cup P_2, f) \leq U(P_2, f).$$

Thus $L(P_1, f) \leq U(P_2, f)$ for all partitions P_1, P_2. Since $L(P_1, f) \leq U(P_2, f)$ holds for every P_2, we have $L(P_1, f) \leq \inf_{P_2} U(P_2, f)$ for every P_1. Taking the supremum over P_1:

$$\sup_{P_1} L(P_1, f) \leq \inf_{P_2} U(P_2, f).$$ □

Theorem 1.12. *A bounded function f is Riemann integrable on $[a, b]$ if and only if for every $\varepsilon > 0$, there exists a partition P of $[a, b]$ such that*

$$U(P, f) - L(P, f) < \varepsilon.$$

Proof. ($\Rightarrow$) Suppose $f \in R$ on $[a, b]$, so $\underline{\int_a^b} f\,dx = \overline{\int_a^b} f\,dx = \int_a^b f\,dx$. Let $\varepsilon > 0$. Since $\int_a^b f\,dx = \sup_P L(P, f)$, there exists a partition P_1 such that

$$\int_a^b f\,dx - L(P_1, f) < \frac{\varepsilon}{2}.$$

Since $\int_a^b f\,dx = \inf_P U(P, f)$, there exists a partition P_2 such that

$$U(P_2, f) - \int_a^b f\,dx < \frac{\varepsilon}{2}.$$

Let $P = P_1 \cup P_2$. Since P refines both P_1 and P_2:

$$U(P, f) - L(P, f) \leq U(P_2, f) - L(P_1, f) = \left(U(P_2, f) - \int_a^b f\,dx\right) + \left(\int_a^b f\,dx - L(P_1, f)\right) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

($\Leftarrow$) Suppose for every $\varepsilon > 0$ there exists P with $U(P,f) - L(P,f) < \varepsilon$. Since $L(P,f) \le \underline{\int} \le \overline{\int} \le U(P,f)$, we have

$$0 \le \overline{\int_a^b} f\,dx - \underline{\int_a^b} f\,dx \le U(P,f) - L(P,f) < \varepsilon.$$

Since this holds for all $\varepsilon > 0$, we conclude $\overline{\int_a^b} f\,dx = \underline{\int_a^b} f\,dx$, so $f \in R$. $\square$

Theorem 1.13. *If $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a,b]$ such that $U(P,f) - L(P,f) < \varepsilon$ and $f \in R$ on $[a,b]$, then for any choice of sample points $t_i \in [x_{i-1}, x_i]$,*

$$\left| \sum_{i=1}^{n} f(t_i)\,\Delta x_i - \int_a^b f\,dx \right| < \varepsilon.$$

Proof. Since $m_i \le f(t_i) \le M_i$ for each i, we have

$$L(P,f) = \sum_{i=1}^{n} m_i\,\Delta x_i \le \sum_{i=1}^{n} f(t_i)\,\Delta x_i \le \sum_{i=1}^{n} M_i\,\Delta x_i = U(P,f).$$

Also, $L(P,f) \le \int_a^b f\,dx \le U(P,f)$. Both the Riemann sum and the integral lie in the interval $[L(P,f),\, U(P,f)]$, which has length less than ε, so

$$\left| \sum_{i=1}^{n} f(t_i)\,\Delta x_i - \int_a^b f\,dx \right| \le U(P,f) - L(P,f) < \varepsilon. \qquad \square$$

Theorem 1.14. *If f is continuous on $[a,b]$, then $f \in R$ on $[a,b]$.*

Proof. Let $\varepsilon > 0$. Since f is continuous on the compact set $[a,b]$, f is uniformly continuous. Thus there exists $\delta > 0$ such that for all $x, t \in [a,b]$,

$$|x - t| < \delta \implies |f(x) - f(t)| < \frac{\varepsilon}{b-a}.$$

Let $P = \{x_0, x_1, \ldots, x_n\}$ be any partition of $[a,b]$ with $\Delta x_i < \delta$ for all i. Then on each subinterval $[x_{i-1}, x_i]$, any two points are within δ of each other, so

$$M_i - m_i < \frac{\varepsilon}{b-a}.$$

Therefore

$$U(P,f) - L(P,f) = \sum_{i=1}^{n} (M_i - m_i)\,\Delta x_i < \frac{\varepsilon}{b-a} \sum_{i=1}^{n} \Delta x_i = \frac{\varepsilon}{b-a} \cdot (b-a) = \varepsilon.$$

By the previous theorem, $f \in R$ on $[a,b]$. $\square$

Theorem 1.15. *If f is monotonic on $[a,b]$, then $f \in R$ on $[a,b]$.*

Proof. Without loss of generality, assume f is monotonically increasing (the decreasing case is analogous). Then on each subinterval $[x_{i-1}, x_i]$,

$$m_i = f(x_{i-1}), \quad M_i = f(x_i).$$

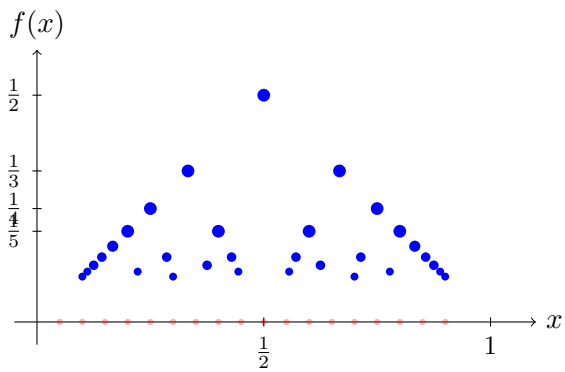
Let $\varepsilon > 0$. Choose the uniform partition $P = \{x_0, x_1, \ldots, x_n\}$ with $\Delta x_i = \frac{b-a}{n}$ for all i, where n is large enough that $\frac{(b-a)}{n}(f(b) - f(a)) < \varepsilon$. Then

$$U(P, f) - L(P, f) = \sum_{i=1}^{n} (M_i - m_i)\,\Delta x_i = \frac{b-a}{n} \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \frac{b-a}{n}\big(f(b) - f(a)\big) < \varepsilon.$$

The sum telescopes, leaving only $f(b) - f(a)$. By the previous characterization, $f \in R$ on $[a, b]$. $\square$

Example. Define $f : [0, 1] \to \mathbb{R}$ by

$$f(x) = \begin{cases} \dfrac{1}{q} & \text{if } x = \dfrac{p}{q} \in \mathbb{Q} \text{ in lowest terms,} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$



Is f Riemann integrable? Yes! Note that $L(P, f) = 0$ for every partition P, since every subinterval contains irrationals. The key observation is that for any $\varepsilon > 0$, only finitely many rationals p/q in $[0, 1]$ satisfy $1/q \geq \varepsilon$, so we can trap them in subintervals of arbitrarily small total length, making $U(P, f)$ arbitrarily close to 0. Thus $\int_0^1 f\,dx = 0$.

Theorem 1.16. *If f is bounded on $[a, b]$ and f has only finitely many points of discontinuity, then $f \in R$ on $[a, b]$.*

Proof. Let $E = \{e_1, \ldots, e_m\}$ be the set of discontinuities of f. Cover E by finitely many disjoint open intervals (u_j, v_j) such that for every $e \in E$, $e \in (u_j, v_j)$ for some j. The complement

$$K = [a, b] \setminus \bigcup_{j=1}^{m} (u_j, v_j)$$

is compact. Since f is bounded, let M be such that $|f(x)| \leq M$ for all $x \in [a, b]$. Then on any subinterval, $M_i - m_i \leq 2M$.

Let $\varepsilon > 0$. Choose $\delta < \dfrac{\varepsilon}{4M}$. Cover E with intervals $[u_j, v_j]$ such that

$$\sum_{j=1}^{m} (v_j - u_j) < \delta.$$

On the compact set $K = [a,b] \setminus \bigcup_j (u_j, v_j)$, f is continuous, hence uniformly continuous. Choose a partition P fine enough that on each subinterval of P lying in K, we have $M_i - m_i < \dfrac{\varepsilon}{2(b-a)}$. Include the points u_j, v_j in P. Then

$$U(P,f) - L(P,f) = \underbrace{\sum_{\substack{i \\ [x_{i-1},x_i] \subseteq K}} (M_i - m_i)\,\Delta x_i}_{< \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \varepsilon/2} + \underbrace{\sum_{\substack{i \\ [x_{i-1},x_i] \subseteq [u_j,v_j]}} (M_i - m_i)\,\Delta x_i}_{\leq 2M \cdot \delta < \varepsilon/2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $f \in R$ on $[a,b]$. □

Theorem 1.17. *Suppose $f \in R$ on $[a,b]$ with $m \leq f(x) \leq M$ for all $x \in [a,b]$, and g is continuous on $[m,M]$. Then $h = g \circ f$ is Riemann integrable on $[a,b]$.*

Proof. Let $\varepsilon > 0$. Since g is continuous on the compact set $[m,M]$, it is uniformly continuous: there exists $\delta > 0$ such that

$$|s-t| < \delta \implies |g(s) - g(t)| < \varepsilon.$$

We may assume $\delta < \varepsilon$. Since $f \in R$, choose a partition $P = \{x_0, \ldots, x_n\}$ such that $U(P,f) - L(P,f) < \delta^2$. Let

$$M_i^f = \sup_{[x_{i-1},x_i]} f, \quad m_i^f = \inf_{[x_{i-1},x_i]} f, \quad M_i^h = \sup_{[x_{i-1},x_i]} h, \quad m_i^h = \inf_{[x_{i-1},x_i]} h.$$

Partition the indices $\{1, \ldots, n\}$ into two sets:

$$A = \{i : M_i^f - m_i^f < \delta\},$$
$$B = \{i : M_i^f - m_i^f \geq \delta\}.$$

For $i \in A$: the oscillation of f on $[x_{i-1}, x_i]$ is less than δ, so by uniform continuity of g, $M_i^h - m_i^h < \varepsilon$.

For $i \in B$: since g is continuous on the compact set $[m,M]$, g is bounded, say $|g| \leq K$. Then $M_i^h - m_i^h \leq 2K$. Also,

$$\delta \sum_{i \in B} \Delta x_i \leq \sum_{i \in B} (M_i^f - m_i^f)\,\Delta x_i \leq U(P,f) - L(P,f) < \delta^2,$$

so $\sum_{i \in B} \Delta x_i < \delta$.

Combining:

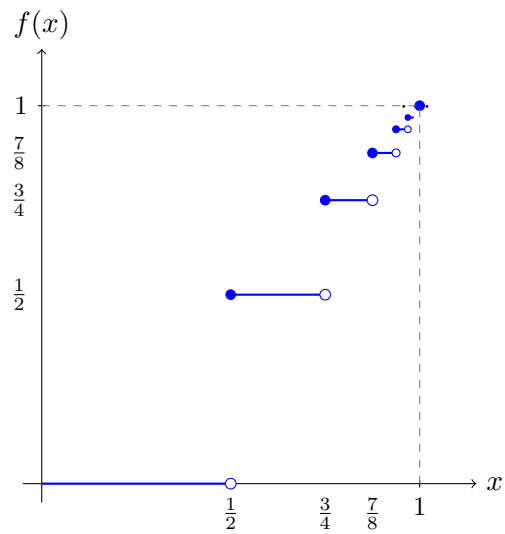
$$U(P,h) - L(P,h) = \sum_{i \in A} (M_i^h - m_i^h)\,\Delta x_i + \sum_{i \in B} (M_i^h - m_i^h)\,\Delta x_i < \varepsilon(b-a) + 2K\delta < \varepsilon(b-a) + 2K\varepsilon.$$

Since ε is arbitrary, $h \in R$ on $[a,b]$. □

Example (Zeno Staircase)**.** Define $f : [0,1] \to \mathbb{R}$ by

$$f(x) = \frac{k}{2^n} \quad \text{for } x \in \left[\frac{k}{2^n}, \frac{k+1}{2^n}\right), \quad \text{i.e., } f = \sum_{n=1}^{\infty} \frac{1}{2^n} \mathbf{1}_{[1-1/2^n,\,1)}.$$

Equivalently, f jumps by $1/2^n$ at $x = 1 - 1/2^n$:



This function is bounded, monotonically increasing, and has countably many discontinuities (at $x = 1 - 1/2^n$ for $n = 1, 2, \dots$). By the previous theorems, $f \in R$ on $[0, 1]$. Note that f has infinitely many discontinuities, so the finite-discontinuity theorem does not apply directly. However, f is monotonically increasing, which guarantees integrability.